

Introduction to Machine Learning Problems Unit

2: Simple Linear Regression

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1. (a) A possible target variable is the students' GPA, as it can be normalized on a scale between 0 and 100.
(b) The target variable is continuous-valued.
(c) The students' previous academic GPA could be a predictor for their current GPA.
(d) A linear model would be reasonable; GPA would directly correlate with the success of a student. Outliers, such as vastly different majors or colleges, would exist, but would not have a severe impact on the linearity of the model.
2. (a) $\bar{x} = 2$ and $\bar{y} = 6$
(b) $s_x^2 = 2.5$, $s_y^2 = 46.5$, and $s_{xy} = 10$
(c) $\beta_0 = -2$, $\beta_1 = 4$, and $RSS(\beta_0, \beta_1) = 26$
(d) $f(x = 2.5) = 8$

3. (a)

$$z(t) \approx z_0 e^{-\alpha t} \tag{1}$$

$$\ln(z(t)) \approx \ln(z_0 e^{-\alpha t}) = \ln(z_0) + -\alpha t \tag{2}$$

z_0 is a constant, and taking the log of the constant results in a constant: $\ln(z_0)$ is our intercept and $-\alpha$ is our slope.

- (b) $\beta_0 = \ln(z_0)$ and $\beta_1 = -\alpha$

- (c) `import numpy as np`

```
y = np.log(z)
```

```
beta_0, beta_1 = np.polyfit(t, y, 1)
```

```
alpha = beta_0
```

```
z_0 = np.exp(beta_1)
```

4. (a)

$$RSS(\beta) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (3)$$

Let $\hat{y}_i = \beta x_i$, then substituting into **Eq.2** results in

$$RSS(\beta) = \sum_{i=1}^n (y_i - \beta x_i)^2 \quad (4)$$

(b)

$$\frac{dRSS(\beta)}{d\beta} = \frac{d}{d\beta} \sum_{i=1}^n (y_i - \beta x_i)^2 \quad (5)$$

$$= 2 * \sum_{i=1}^n (y_i - \beta x_i)(-x_i) \quad (6)$$

$$= -2 * \sum_{i=1}^n (x_i y_i - \beta x_i^2) \quad (7)$$

Setting the equation to 0 results in:

$$-2 * \sum_{i=1}^n (x_i y_i - \beta x_i^2) = 0 \quad (8)$$

$$\sum_{i=1}^n (x_i y_i - \beta x_i^2) = \quad (9)$$

$$\sum_{i=1}^n x_i y_i = \beta \sum_{i=1}^n x_i^2 \quad (10)$$

Solving for β results in

$$\beta = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$$